

exercise4.py

castleupdate...

https://castleupdate-dynastygossip-3000.codio.io

Connected (encrypted) to castleupdate-dynastygossip:0

Guide

Send CtrlAltDel

libopenjp2.so.7libstdc++.so.6libpcre.so.1libpng16.so.16libpython3.10.so.1.0libreadline.so.8libSDL2-2.0.so.0libtiff.so.5libtinfo.so.6libwebpdemux2.so.8libwebpmux3.so.0

Use the link below to view the

Terminal

```
File "logging/__init__.py", line 1506, in error
File "logging/__init__.py", line 1624, in _log
File "logging/__init__.py", line 1634, in handle
File "logging/__init__.py", line 1696, in callHandlers
File "logging/__init__.py", line 968, in handle
File "logging/__init__.py", line 1108, in emit
Message: 'Input: MTDev is not supported by your version of linux'
Arguments: ()
[INFO ] [Window ] Provider: sdl2
[INFO ] [GL      ] Using the "OpenGL" graphics system
[INFO ] [GL      ] Backend used <sdl2>
[INFO ] [GL      ] OpenGL version <b'3.1 Mesa 20.0.8'>
[INFO ] [GL      ] OpenGL vendor <b'VMware, Inc.'>
[INFO ] [GL      ] OpenGL renderer <b'llvmpipe (LLVM 10.0.0, 128 bits)'>
[INFO ] [GL      ] OpenGL parsed version: 3, 1
[INFO ] [GL      ] Shading version <b'1.40'>
[INFO ] [GL      ] Texture max size <8192>
[INFO ] [GL      ] Texture max units <32>
[INFO ] [Window ] auto add sdl2 input provider
[INFO ] [Window ] virtual keyboard not allowed, single mode, not docked
[INFO ] [Base    ] Start application main loop
[INFO ] [GL      ] NPOT texture support is available
[INFO ] [Base    ] Leaving application in progress...
(exercise4) codio@castleupdate-dynastygossip:~/workspace/ex4$
```

You should see the following output:

Exercise4

Hello

from

Kivy

pyinstaller exercise4.py --name hello -w

Submit your code for evaluation.

Here is one possible solution for the Python code:

```
from kivy.app import App
from kivy.uix.gridlayout import GridLayout
from kivy.uix.label import Label

class Exercise4(App):
    def build(self):
        # define the window
        self.window = GridLayout(cols=3)

        # define the widgets
        label1 = Label(text='Hello', font_size=36)
        label2 = Label(text='from', font_size=36)
        label3 = Label(text='Kivy', font_size=36)

        # add the widgets
        self.window.add_widget(label1)
        self.window.add_widget(label2)
        self.window.add_widget(label3)

        return self.window

if __name__ == "__main__":
    Exercise4().run()
```

- Import the Kivy packages for `App`, `GridLayout`, and `Label`
- Create the `Exercise4` class and inherit from `App`
- Create a grid layout with 3 columns
- Create three separate labels
 - The text in the labels should be `'Hello'`, `'from'`, and `'Kivy'`
 - The font size for each label should be 36
- Add the labels to the window; be sure to add them in the proper order
- Return the window
- Call the `run()` method on the Kivy app when the script is executed

Here is the terminal command to create the Kivy GUI app:

pyinstaller exercise4.py --name hello -w